

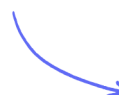
Multiple Choice Questions

Introduction to Kinetics & Equilibria

Collision Theory / Rates of Reaction / Maxwell-Boltzmann Distributions / Catalysts & Energy / Dynamic Equilibrium in Reversible Reactions / Le Chatelier's Principle / Industrial Compromises

Easy (1 question)	/1
Medium (4 questions)	/4
Total Marks	/5

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Easy Questions

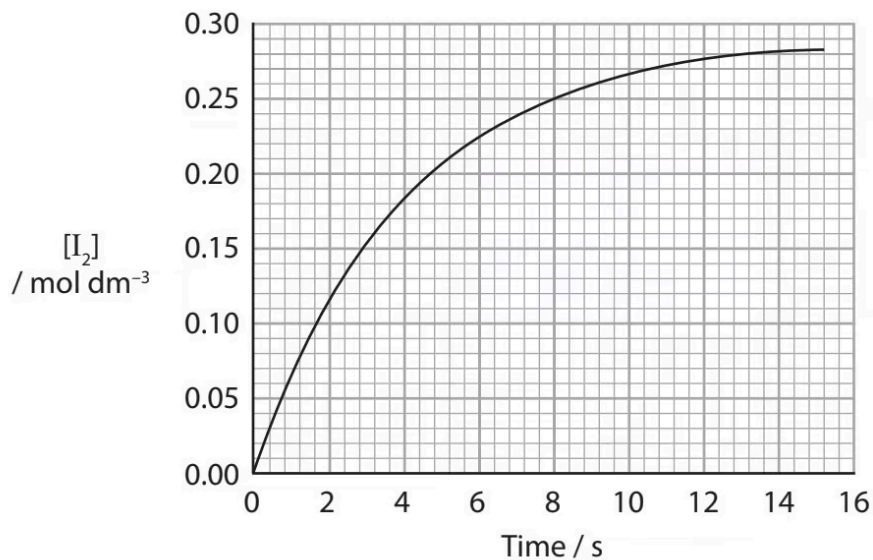
- 1 What is the effect of increasing temperature on the average energy of the particles in a reaction and on the activation energy of the reaction?

		Effect on the average energy	Effect on the activation energy
☐	A	unchanged	decreased
☐	B	unchanged	unchanged
☐	C	increased	decreased
☐	D	increased	unchanged

(1 mark)

Medium Questions

- 1 The graph shows how the concentration of iodine changes with time in a reaction.



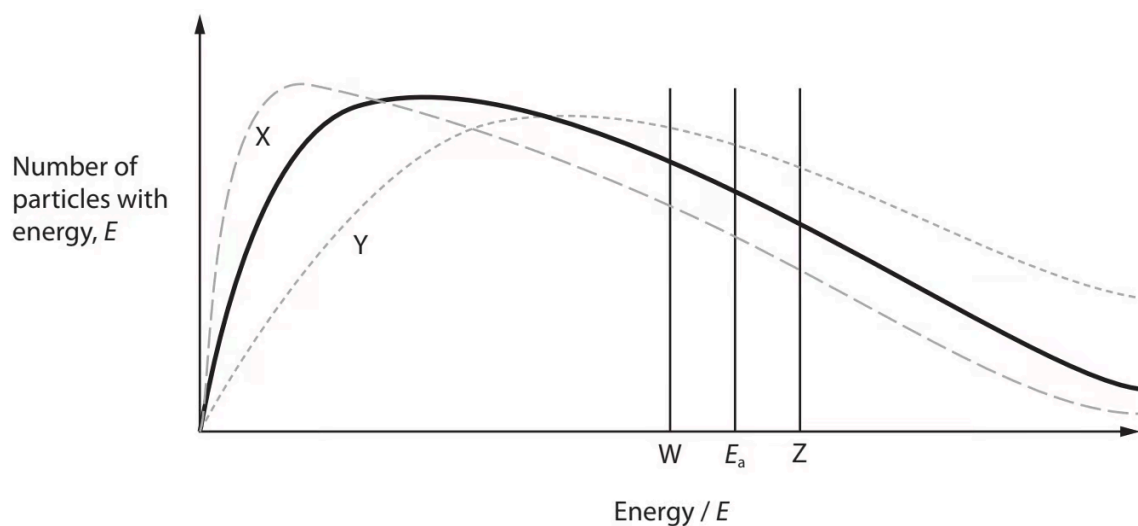
What is the value for the rate of reaction, in $\text{mol dm}^{-3} \text{ s}^{-1}$, at 8 seconds?

- A. 0.01
- B. 0.02
- C. 0.03
- D. 0.25

(1 mark)

- 2 The solid line on the graph below shows the Maxwell-Boltzmann distribution for an

uncatalysed reaction. E_a is the activation energy of this reaction.



Which row shows the correct Maxwell-Boltzmann curve and activation energy for the reaction at a higher temperature with a catalyst?

		Maxwell-Boltzmann curve	Activation energy
☐	A	X	W
☐	B	X	Z
☐	C	Y	W
☐	D	Y	Z

(1 mark)

3 Which conditions give the highest yield for the forward reaction?



- A. high temperature, high pressure
- B. high temperature, low pressure
- C. low temperature, high pressure
- D. low temperature, low pressure

(1 mark)

4 Nitrogen dioxide and dinitrogen tetroxide exist in equilibrium.



When an equilibrium is set up in a gas syringe, the mixture is pale brown.

When the mixture is compressed the colour becomes

- A. darker
- B. lighter
- C. darker and then lighter
- D. lighter and then darker

(1 mark)